
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture of Experts (MoE) models are now a common route for scaling large language models, but the representational consequences of replacing dense computation with routed experts are still not well characterized. Prior work shows that dense models can use *superposition* to represent more features than dimensions, with feature sparsity and feature importance shaping when that packing occurs. We show that this dense-model account does not transfer directly to MoEs: in our controlled MoE setting, feature sparsity and feature importance do not induce the same discontinuous phase changes, while network sparsity (the ratio of active to total experts) more consistently explains representational behavior. We introduce metrics for measuring superposition across experts and find that higher network sparsity is associated with greater *monosemanticity*. We also frame expert specialization in terms of monosemantic feature representation rather than only load balancing, showing that experts can organize around coherent feature combinations when the router is initialized to make such structure accessible. Within these toy reconstruction settings, the results indicate that conditional computation can broaden the design space for interpretable MoEs without requiring an immediate performance sacrifice, qualifying the view that interpretability and capability must always trade off.

1 INTRODUCTION

Mixture of Experts (MoEs) are widely used in contemporary language-model families, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), because they offer a general mechanism for increasing parameter count while controlling active computation (Shazeer et al., 2017; Fedus et al., 2022). Follow-up work has improved both routing (for example, Expert-Choice routing) and training stability or transfer (for example, ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Theory and empirical evidence also suggest that learnable routers can recover latent cluster structure in data, offering one explanation for why experts may specialize (Dikkala et al., 2023). Yet even as MoEs become a standard architectural option, their internal representations remain comparatively underexplained from a mechanistic interpretability perspective.

Recent interpretability-motivated MoE designs show that expert behavior can be made more legible through architectural choices. Yang et al. (2025b) propose MoE-X, encouraging sparsity-aware routing and using wide, ReLU-based experts to reduce polysemanticity, while Park et al. (2025) introduce MONET, which scales the expert count to support capability editing through expert activation. These approaches demonstrate useful intervention points, but they leave a broader mechanistic question open: how does routing itself change feature representation, superposition, and specialization when no additional interpretability regularizer is imposed? Consistent with this gap, Mu & Lin (2025) identify mechanistic interpretability as a key open challenge for MoE research.

A central obstacle for interpreting many neural architectures is superposition: models can represent more features than they have dimensions. This packing is capacity-efficient for sparse features, but it makes individual neurons polysemantic and therefore weakens the connection between a unit-level observation and a single human-interpretable concept.

MoE architectures add a second kind of sparsity to this representational problem: network sparsity. Dense models apply the same computation to every input, whereas MoEs activate only a subset of their parameters (Shazeer et al., 2017). As a result, a MoE need not respond to feature sparsity only by packing features into shared dimensions; it can also route different regions of the feature space to different experts, potentially reserving expert capacity for recurring feature combinations.

We therefore ask whether (1) MoEs exhibit less superposition than comparable dense models, (2) the dense-model phase changes induced by feature sparsity and relative feature importance also appear inside MoE experts, and (3) expert specialization can be understood as a property of feature representation rather than only as a property of aggregate load balancing.

We study these questions in simple reconstruction models that extend Elhage et al. (2022)’s toy framework to routed experts. The contributions are: (1) in these MoE settings, the sharp dense-model phase changes are replaced by more continuous transitions as network sparsity increases; (2) MoEs exhibit greater monosemanticity (less superposition) than dense models under matched active- and total-parameter comparisons, with cleaner feature representations inside individual experts; (3) we propose an interpretability-focused definition of expert specialization based on monosemantic feature representation, and show that experts can organize around coherent feature combinations rather than merely satisfying arbitrary load-balance constraints.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis frames concepts as directions in activation space (Park et al., 2024), but across many settings the number of useful or interpretable features can exceed the number of available dimensions. Elhage et al. (2022) formalized this mismatch as superposition, showing in dense toy models that non-orthogonal feature packing increases capacity while producing polysemantic representations. Although superposition can be theoretically efficient (Scherlis et al., 2025), practical interpretation often requires post-hoc disentanglement tools such as Sparse Autoencoders to recover more monosemantic features (Bricken et al., 2023). Other work has studied how data correlations shape superposition (Prieto et al., 2025) and how interference patterns emerge or can be mitigated (Gurnee et al., 2023). We extend this line of inquiry from dense feature spaces to routed expert architectures, where network sparsity can govern representational strategies in addition to feature sparsity.

Interpretability of MoEs. MoEs are a major scaling tool for language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), yet their mechanistic interpretation is less developed than for comparable dense architectures. Existing analyses often operate at a macroscopic level, examining routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), or latent cluster discovery (Dikkala et al., 2023). More recent interpretability-focused systems encourage specialization through constraints such as pre-defined concept routing (Yang et al., 2025b) or expert counts tied to vocabulary size (Park et al., 2025). These directions are useful for editing and control, but they do not fully explain the feature geometry that routed experts learn by default. We address this gap by analyzing how experts change superposition, showing greater monosemanticity than dense baselines in our matched toy settings, and proposing a feature-based account of expert specialization.

3 BACKGROUND

A primary goal of mechanistic interpretability is to reverse engineer neural networks by decomposing their representations into human-interpretable concepts, often called ‘features’. In the linear feature view, a hidden state h can be approximated as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ denotes the direction associated with feature i , α_i is its activation strength (roughly, how strongly feature i is present in the input), and F is the full set of represented features.

The *superposition hypothesis* states that models can represent far more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). To fit many features into a lower-dimensional latent space, feature vectors in F cannot all remain orthogonal. When two features interfere, $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we describe them as being in *superposition*. This tradeoff can be acceptable when features are sparse, as is often assumed for text-like domains, but it reduces interpretability because α_i can include activation caused by other features rather than by feature i itself.

Monosemanticity is usually defined at the neuron level: a neuron’s activation cleanly tracks a single α_i when features are basis-aligned. If features are orthogonal but not basis-aligned, individual neurons may still be *polysemantic* even though feature interference is low. Our analysis therefore targets reduced superposition rather than strict basis alignment. When we call a model, expert, or feature “more monosemantic,” we use the phrase as shorthand for exhibiting less superposition under these metrics.

4 DEMONSTRATING SUPERPOSITION

MoEs are often described as collections of dense subnetworks, suggesting that each expert might behave like a dense model restricted to a subset of inputs. That intuition is useful but incomplete: routing may change the feature geometry each expert learns. We test this by comparing superposition in matched dense and MoE toy models.

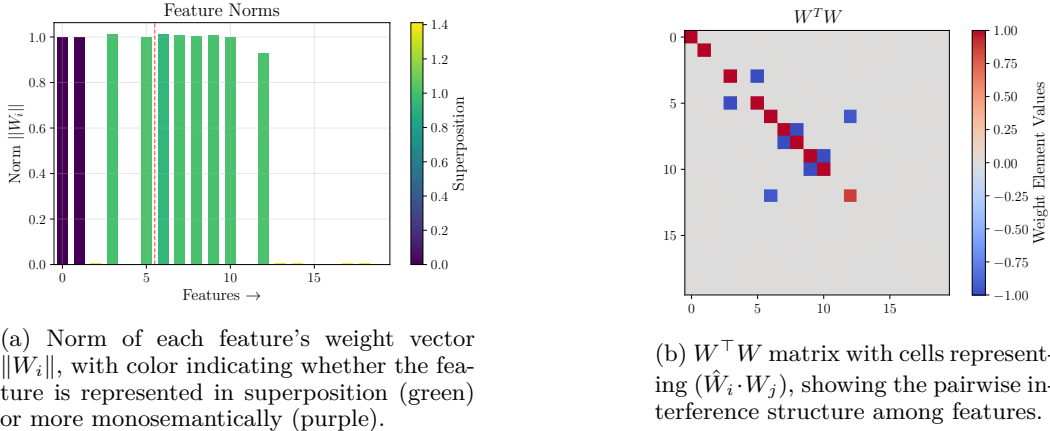


Figure 1: Feature representation and superposition in a dense baseline with $n = 20$ features and $m = 6$ hidden dimensions, using importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. Color reports the superposition score $\sum_j (\hat{W}_i \cdot W_j)^2$.

4.1 EXPERIMENTAL SETUP

Our experiments ask how a routed model projects a high-dimensional vector, $x \in \mathbb{R}^n$, into lower-dimensional expert representations, $h \in \mathbb{R}^m$, and then reconstructs the original input. This keeps the controlled reconstruction setup of Elhage et al. (2022) while extending it from a single dense representation space to multiple routed expert spaces.

The input distribution. The input vector x represents activations from an idealized disentangled source, where each dimension x_i is a distinct independent feature—that is, a perfectly neuron-aligned and monosemantic representation. We sample $x_i \sim U(0, 1)$, except that for sparsity $S \in [0, 1]$, $P(x_i = 0) = S$. Because features need not contribute equally to the objective, each feature x_i receives a scalar importance I_i that weights reconstruction loss. To isolate phase-transition behavior for one feature, we also scale the magnitude of the

final feature by $r \in \mathbb{R}^+$, giving $I = (1, 1, \dots, r)$. Together, I_i and r let us vary feature signal strength while keeping the underlying task and distribution family fixed.

Model Architecture. The MoE has E experts, with each expert e parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. A learned router $g(x) = \text{softmax}(W^r x)$, where $W^r \in \mathbb{R}^{E \times n}$, assigns each input to its top- k experts. Each active expert maps the input to a lower-dimensional hidden state $h^e = W^e x$ and produces the reconstruction $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ ¹. The model output is the weighted sum over active experts, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, with w_e denoting the renormalized gating weights.

All models are trained with an L2 reconstruction objective weighted by feature importances, I_i , namely $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, where expert collapse would obscure the phase diagrams, we additionally use the standard auxiliary load-balancing loss (Fedus et al., 2022), $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Here, N is the total number of experts, f_e is the fraction of samples in a batch routed to expert e , P_e is the average gating probability assigned to expert e across the batch, and $\alpha = 0.01$ sets the penalty strength. We omit auxiliary load balancing in Section 4 so that the dense-MoE comparison reflects the architectural effect of routing on superposition rather than an explicit balancing regularizer.

4.2 MEASURING FEATURE CAPACITY

To compare representations across architectures, we measure two broadly useful feature-level quantities: *representation strength* and *interference*. The norm $\|W_i^e\|$ of feature i 's weight vector in expert e indicates how strongly that feature is represented by the expert: $\|W_i^e\| \approx 1$ when the feature is fully represented and near zero when it is not learned. We visualize how feature i interferes with other features in expert e using the Gram matrix $W^\top W$, whose off-diagonal entries capture pairwise interference.

Figures 1a and 2a show that, under equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$), the dense model and MoE represent a comparable number of features (10 vs 8) with similar norm scales. The difference is not simply how many features are learned, but how interference is distributed. The MoE experts still show local superposition (for example, Expert 0 in Fig 2b), yet the global interference structure is partitioned: unlike the dense model in Figure 1b, where any feature can interfere with any other, features routed to different experts do not interfere. In this setting, MoEs allocate capacity by partitioning the feature space, reducing the global scope over which interference can occur.

Expert Feature Dimensionality. To characterize how routed models allocate limited representational capacity, we use feature dimensionality, the “fraction of a dimension” assigned to a particular feature (Elhage et al., 2022). For feature i in expert e , we define dimensionality as

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e ranges from zero, when the feature is not learned, to one, when it is represented monosemantically. The aggregate representational capacity of a MoE is therefore $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. A representation is efficiently packed when feature dimensionalities sum to the number of embedding dimensions, i.e. $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For MoEs, the corresponding relation is $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint, indicating that routed experts can pack features as efficiently as dense models with the same total parameter budget while changing where interference occurs.

¹The ReLU at the output ensures non-negative reconstructions, which matches our input distribution where $x_i \in [0, 1]$ when nonzero. Furthermore, the off-diagonal terms in $(W^e)^\top W^e$ create negative interference and ReLU suppresses these negative components. When features are sparse, negative interference becomes effectively “free” as it is filtered to zero, incentivizing model configurations with negative off-diagonal terms (e.g., antipodal pairs).

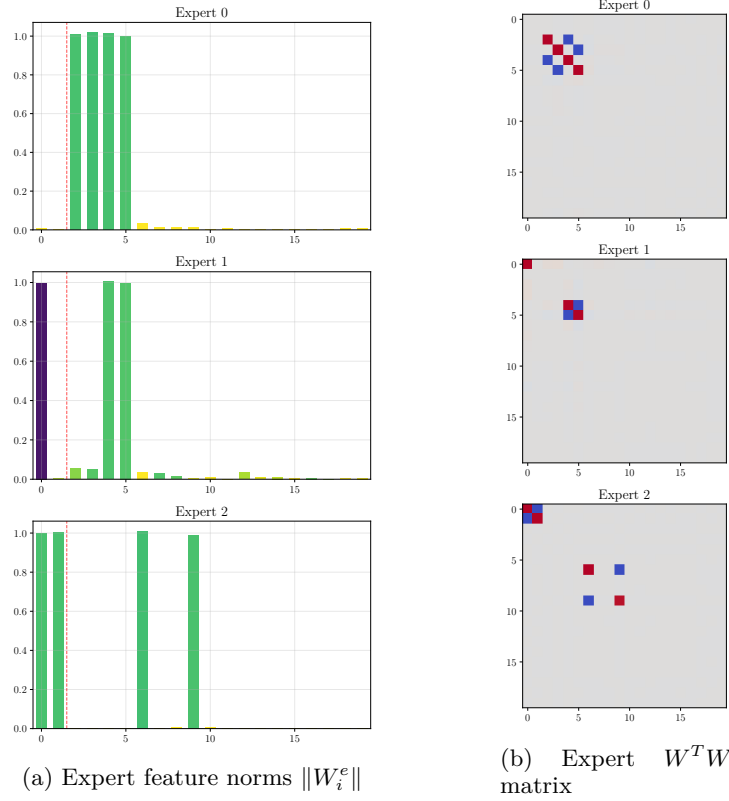


Figure 2: Feature representation and superposition in a routed MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert (top- $k = 1$ routing), using importance $I = 0.7^i$ and feature density $1 - S = 0.1$.

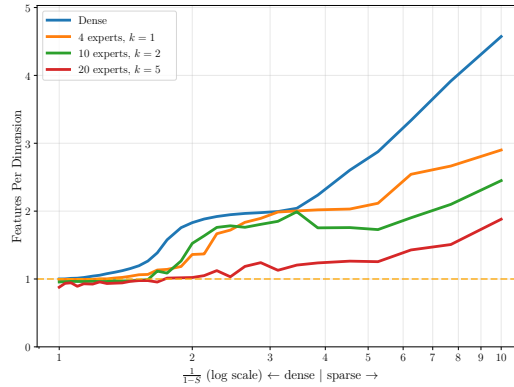


Figure 3: Features per dimension versus inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures with uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the highest superposition, while superposition decreases as the number of experts increases: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); 20 experts with $m = 1$, $k = 5$ (red). All models have equal total parameters and similar k/E . The dashed line at 1.0 marks monosemantic representation.

Features per Dimension & Network Sparsity. Because both dense and MoE models efficiently pack features, we compare them using the number of features per dimension. This quantity measures superposition across architectures and shows how expert count changes

representational geometry at different feature sparsities. Values greater than one indicate superposition, since the model is representing more features than available dimensions. For a MoE, we define features per dimension as

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the Frobenius norm and p_e is the expected usage probability of expert e across a batch, equivalently the average renormalized gating weight after top- k routing.

For MoEs with the *same* total number of parameters as the dense model, the dense model has more features per dimension (Figure 3), and thus more superposition. The dense model uses superposition to represent more learned features than total dimensions ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE tends to stay closer to the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). As the number of experts increases while total parameters and the ratio k/E remain roughly fixed, features per dimension decreases. In these settings, increasing the number of experts makes representations more monosemantic. The dense model’s greater superposition also gives it consistently lower reconstruction loss than the MoEs in Figure 6 of Appendix A.1, with a loss gap of $\sim 0.03 - 0.08$ at a given sparsity. However, the MoE loss becomes consistently comparable as expert count increases. Appendix A.2 gives theoretical intuition for this pattern.

5 PHASE CHANGE

The preceding results suggest that MoEs can learn a similar number of features as dense models while distributing those features across experts with less global interference. This shifts the key question from how many features are learned to how feature allocation changes under routing. We therefore examine whether input-distribution properties such as feature sparsity and importance still induce the physics-inspired *phase changes* observed in dense representations.

With finite capacity, a model has only a few qualitative options for any feature: ignore it, represent it in superposition, or represent it more monosemantically. A phase change occurs when the boundary between these regimes is discrete as feature sparsity and relative importance vary.

Dense toy models show discontinuous ‘phase changes’ between internal representational strategies (Elhage et al., 2022). Varying feature sparsity and relative feature importance can move a dense model between regimes, with higher feature sparsity often making superposition more favorable. By constructing analogous phase diagrams for individual MoE experts, we can test whether routed architectures follow the same dense-model regimes or use a broader set of allocation strategies.

Setup. We use the setup from Section 4.1, adding load-balancing loss in the phase-change experiments to avoid specialization collapse, where the router and experts fall into local minima that ignore some experts entirely. We study three architectures, all with one active expert ($k = 1$): (A) $n=2$, $m=1$; (B) $n=3$, $m=1$; and (C) $n=3$, $m=2$. For each architecture, we report expert-specific phase diagrams across feature sparsity and last-feature relative importance while increasing network sparsity by raising the expert count (E) up to the number of input feature dimensions (n).

Here we match active rather than total parameters, setting $m_{\text{dense}} = km_{\text{moe}}$ and ignoring router parameters. This comparison isolates representational differences among architectures with the same active capacity; otherwise, changes could be attributed to active-parameter count rather than routing. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2. However, the latter uses only one hidden dimension ($m = 1$) to encode the same number of input features ($n = 3$) that the former encodes with two hidden dimensions ($m = 2$), so specialization cannot be interpreted through superposition alone.

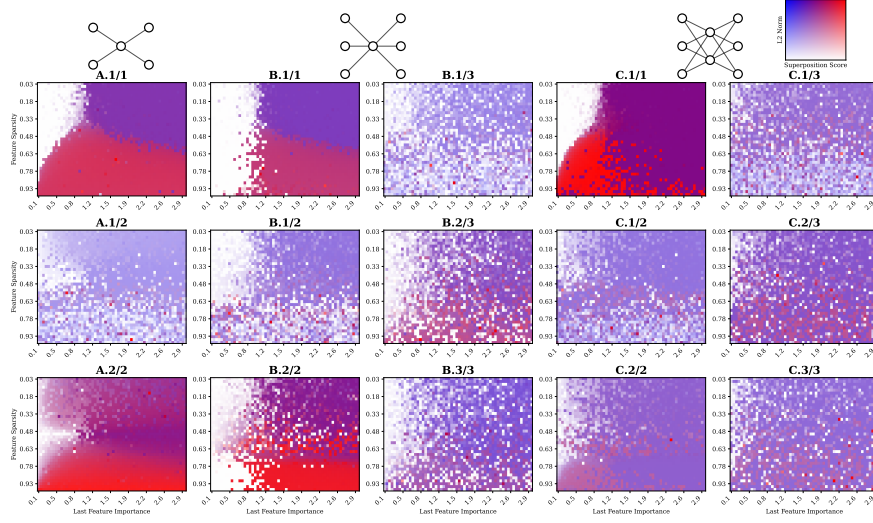


Figure 4: For each expert and input dimension (feature), the diagram shows whether the last feature is ignored (white), represented more monosemantically (blue-purple), or represented in superposition (red). We plot joint feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j<n}(\hat{W}_n \cdot W_j)^2$) over feature sparsity $S \in [0,1]$ and relative last-feature importance $I_n \in [0.1,3]$, where n denotes the last feature among n total features. For each cell, we train ten models and select the lowest-loss run, using load-balancing loss throughout this section. Low L2 norm ($\|W_n\|$) appears white, indicating that the model ignores the last feature; otherwise, low superposition appears blue-purple and high superposition appears red. Cell (i,j) in subfigure X.e/E denotes expert e of E total experts trained on architecture X with last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 denotes the dense model.

Results and Takeaways. All single-expert, dense cases show a clear phase change (Figure 4.X.1/1), matching Elhage et al. (2022). As the number of experts increases, these discrete boundaries disappear or soften. Some $E = 2$ MoE experts still resemble their corresponding dense diagrams (Figure 4.X.2/2), but the transitions are more continuous. In each setting, one expert becomes more monosemantic and specializes in the feature with higher relative importance. Experts that differ most from the dense diagrams have uniformly lower superposition scores (more blue), again indicating more monosemantic representations. This supports the previous section’s broader pattern: routed MoEs favor lower superposition than dense counterparts under the tested conditions.

In the $n = 2, m = 1$ setting (Figure 4.A), the dense model ignores the last feature when feature sparsity is low, whereas the comparable MoE preserves that feature more often because routing provides additional allocation capacity. With three input dimensions (Figure 4.B), this effect does not appear: experts are already superimposing the other two features, leaving no room for the third feature in a single hidden dimension. Unlike the other cases, architecture B lacks enough hidden capacity, even with superposition in the high-sparsity regime, to represent all features. Still, the MoE does not show a clean dense-style phase change, except for a mostly discrete but mixed region around feature sparsity 0.5–0.7 in 4.B.2/2. For $m = 2$, the dense model’s white region in Figure 4.C.1/1 corresponds to ignoring the last feature in the mid- to low-feature-sparsity regime when that feature is relatively less important. As network sparsity increases across the other Figure 4.C models, the last feature is represented with greater L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). Thus, dimensionality in the low-relative-importance region increases with network sparsity, consistent with Figure 3.

Across Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2, feature sparsities from roughly 0.48 to 0.7 produce a mixed region containing polysemantic representation, monosemantic representation, and feature omission. This suggests that MoEs can occupy an intermediate regime in which polysemantic and monosemantic solutions have comparable loss, making the model

less likely to commit consistently to the strategies seen at low or high feature sparsity. Dense models do not show the same pattern, further indicating that routed experts learn distinct representational strategies.

Conclusion. All experiments in this section use $\text{top-}k = 1$, so only one expert is active for each input. Even under this dense-like active-computation constraint, the $E = 2$ cases can behave very differently from a dense model, including when superposition would provide enough hidden capacity to represent all features. This makes it misleading to treat MoEs as simple aggregations of dense models. By restricting each expert to a routed subset of the feature domain, the router substantially changes expert learning and representation.

6 EXPERT SPECIALIZATION

Having shown that MoEs exhibit less superposition in these settings, we next ask how the resulting more monosemantic features are organized within experts and how that organization relates to specialization.

Expert specialization in MoEs is often operationalized through load balancing across experts and inputs (Chaudhari et al., 2025). Load balancing is important, but by itself it misses a broader semantic intuition: a specialized expert should be used preferentially when the concepts it represents are present in the input.

We therefore define an expert as specialized when it *occupies* particular feature directions in input space and represents those features relatively monosemantically. We show that these two conditions are correlated in our toy models and that, when they hold, they also encourage balanced expert usage as a consequence rather than as the sole definition of specialization.

Because we fix $k = 1$, the router partitions feature space into convex cone regions (Appendix A.3), with each region assigned to one expert. Thus $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points in the cone and s is any positive scalar. If a feature vector $\vec{x}$ is routed to expert e , every positive rescaling $s\vec{x}$ is routed to the same expert. We say in this case that feature x lies in expert e ’s routed region, or that expert e *occupies* x .

Empirically, small models that distribute the input space across more experts tend to achieve lower loss (Appendix A.4). This motivates a more general question: when the router allocates a region of input space to an expert, does that allocation predict the features the expert represents? Our specialization definition predicts a link between routed occupancy and *monosemanticity*, and we test this correlation in larger toy models such as $m = 10$.

For models with $n > 2$, we first test whether initializing experts to occupy particular input-space features makes those experts more monosemantic with respect to those features. We then test the converse: for features that an expert ultimately represents monosemantically, we ask whether the same expert occupies those features in the input space.

When the gate matrix is initialized with ones along the main diagonal, each expert monosemantically represents the single feature it initially occupied, and no other feature, as shown in Figure 5a. Under ordered k -hot router initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b). The remaining experts, which are initialized over less important features, do not monosemantically represent those less important features, nor do they take over the five most important features outside their initial routed regions. When feature-importance order is broken and the features initially occupied by each expert are randomized, each expert monosemantically represents only the most important feature or features in its initial routed set (Figure 5c).

These results show a strong relationship between the features initially routed to an expert and the features that expert later represents monosemantically. They also show that importance still matters: experts monosemantically represent important features, whether the initialization assigns each expert one important feature directly or gives the expert a set from which it effectively selects the most important feature to represent cleanly.

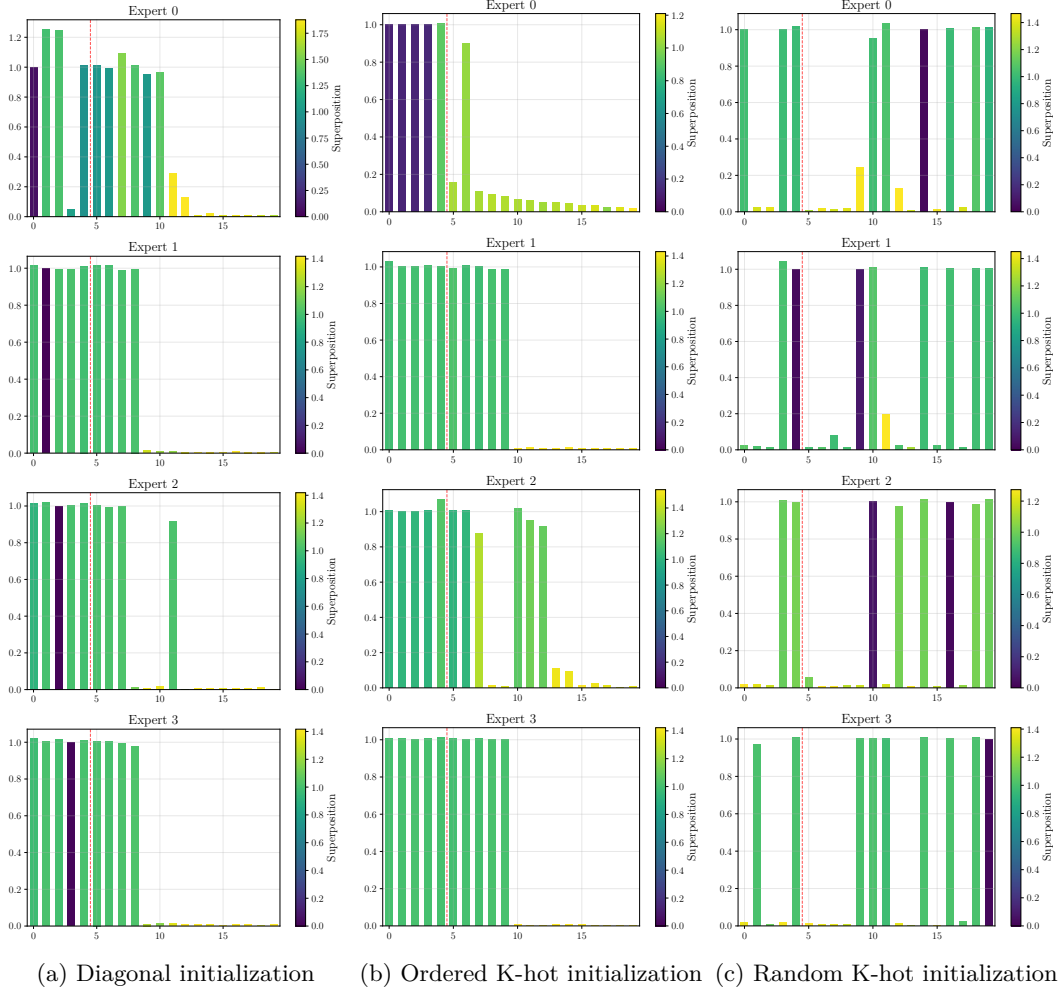


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) under three router initialization schemes, with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for that dimension), while relative feature importance decreases exponentially from feature one to feature 20. In (b), the gate matrix uses an “ordered k-hot” initialization: the first expert aligns with the first five features, and each later expert aligns with the next five features. Relative feature importance is the same as in (a). In (c), the gate matrix uses a “random k-hot” initialization, assigning each expert five random features so that experts share no feature but collectively cover all 20 features. Relative feature importance still decreases exponentially, but its ordering is randomly distributed across features.

With uniform feature importance, experts place all features in superposition and interference increases. Even in this less differentiated setting, the features initially occupied by an expert remain *relatively* more monosemantic, with superposition scores 1.03 standard deviations below that expert’s average.

We next test whether experts that represent features monosemantically also occupy those features in the input space. We measure usage when the relevant features are *one of many* active features and when they are the *only* active features. The latter case directly estimates the probability that the expert occupies those feature directions. Table 1 shows that the correlation holds for both xavier and k-hot initialization. With $E = 10$, mean usage near $\sim 10\%$ corresponds to even load balancing, and in every case expert usage rises substantially when the expert’s monosemantic feature or features are active. When those features are the

only active features, the corresponding expert dominates routing. Under k-hot initialization, 100% of features represented monosemantically by an expert are also occupied by that expert.

Table 1: Monosemantic feature and usage statistics per expert for $n = 100, m = 10, E = 10$. We train one hundred models for each initialization scheme (xavier and k-hot), yielding 1000 experts per scheme. Statistics are aggregated across models after classifying experts by the number of features they represent monosemantically. For the feature(s) represented monosemantically by an expert, we report expert usage when those feature(s) are among several active input features and when those feature(s) are the *only* active input features.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	–	–	–
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	–	–	–
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	–	–	–
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

Experts that represent more features monosemantically are specialized in a stronger feature-based sense. Their average usage on arbitrary inputs may decrease, but their conditional usage rises sharply when their specialized features are active. This pattern holds in all reported cases except the xavier-initialized model with one four-feature monosemantic expert, where utilization drops substantially.

7 CONCLUSION

We studied how routing through experts changes superposition in MoEs. Across the controlled settings in this paper, MoEs exhibit greater monosemanticity than dense networks and do not reproduce the same sharp dense-model phase changes. We also introduced a feature-based definition of expert specialization and showed that experts can organize around coherent feature directions when initialization makes that organization accessible. Because the evidence comes from simple autoencoder toy models with synthetic data, the extension to large-scale transformers with unknown feature distributions remains open (see Appendix A.6). Within those limits, the results show that toy MoEs can approach dense-model loss while maintaining more interpretable representations, suggesting that conditional computation may relax, rather than simply sharpen, the tension between mechanistic interpretability and capability. Future work should identify which training conditions favor monosemanticity in MoEs, how MoE training dynamics differ from dense-model dynamics, and when specialization emerges in larger and more realistic systems. Answering these questions could inform the design of language models that are both performant and more mechanically transparent.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

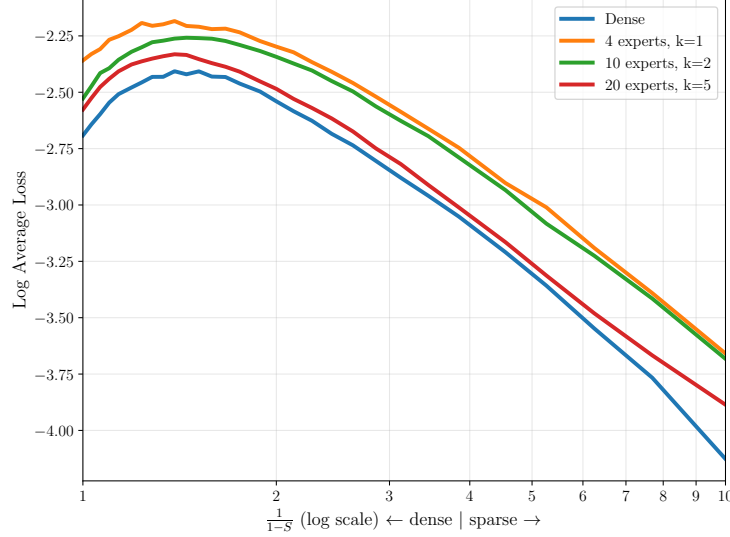


Figure 6: Log average loss versus feature density ($\frac{1}{1-s}$) for dense ($m = 20$) and MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all using uniform feature importance ($I_i = 1.0$) and $n = 100$ input features. Results are averaged over five runs per sparsity level. The dense model has lower loss than all MoEs at every sparsity level, but MoE loss approaches the dense loss as expert count increases.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition exploits the mismatch between sparse feature activation and dense computation by compressing rarely active features into shared dimensions. MoEs address the same mismatch structurally through conditional computation. By aligning activation sparsity with routed expert usage, MoEs reduce the computational penalty of dedicating capacity to rarely active features. As Elhage et al. (2022) note, when computation is spent only on active features, replacing polysemantic neurons with dedicated monosemantic ones can become preferable, trading the *compression* of superposition for the *selection* of routing.

Geometrically, this structural change appears as reduced *interference*. Even when the global ratio of represented features to parameters is fixed, the router partitions feature space so that a feature routed to expert e competes only with the subset of features assigned to that expert. Interference is therefore governed by the local expert matrix $(W^e)^\top W^e$ rather than the dense model’s global $W^\top W$. This partitioning reduces the number of possible interfering features for any given feature vector, weakening the optimization pressure to pack features in superposition and making monosemantic expert representations more accessible.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

When $k = 1$, the router $g(x) = \text{softmax}(W^r x)$ induces the same expert choice as $\text{argmax}(W^r x)$. The region routed to expert i is described by $\forall j \neq i, (w_i - w_j)^\top x > 0$, where w_i and w_j are rows of W^r . These constraints are homogeneous linear inequalities, and regions defined by such inequalities are convex cones. If x satisfies the inequalities, any positive rescaling by s also satisfies them. Likewise, if x_1 and x_2 satisfy them, then $x = x_1 \lambda + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$ also satisfies them.

For $k > 1$, the set of inputs routed to a particular expert e becomes a union of convex cones. A union of convex cones is not generally a convex cone, so the feature-occupancy interpretation used here may not extend directly beyond the $k = 1$ setting.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

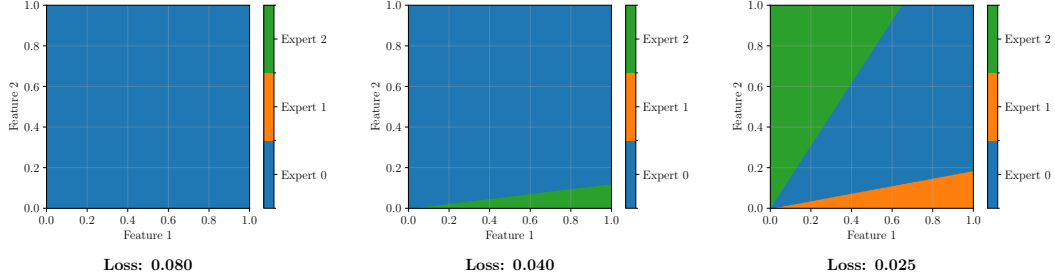


Figure 7: Expert routing for three identical models with different initialization schemes, using $n = 2$, $E = 3$, $m = 1$. The first model (left) has the highest loss (0.08) and routes all inputs to one expert. The second model (middle) has lower loss (0.04) and routes a small input region, specifically cases where feature 1 is active, to a second expert. The third model (right) has the lowest loss (0.025) and distributes the input space across all experts: one expert is selected when only feature 1 is active, one when only feature 2 is active, and one when both are active.

In small models ($n = 2$, $m = 1$, $E > 1$), tested with $E \in [2, 7]$, lower loss is associated with distributing the input space across more experts. Keeping $n = 2$ makes the routed regions directly visualizable, as shown in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the single-layer, single-nonlinearity, top- $k = 1$ toy MoEs studied here, a theoretical construction maps a dense model to a monosemantic MoE when the number of active features is bounded by a sparsity constraint.

Assume that every input has at most a active features, so $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Also assume that this bound is no larger than an expert’s hidden dimensionality m , and that m is no larger than the total number of input features n ($a \leq m \leq n$). The monosemantic MoE construction assigns an expert to each subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$, so that the expert preserves that subset of active features monosemantically. (Equivalently, it is sufficient to take only subsets with $|S| = a$.) The router then selects the expert corresponding to the active feature set, whose size is bounded by a :

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Because $|S| \leq a \leq m$, each expert has enough capacity to represent its assigned active features without superposition. Thus, under the active-feature bound, every possible active feature combination can receive a dedicated expert with sufficient monosemantic capacity. The number of possible experts needed is $\binom{n}{m}$.

This theoretical MoE achieves zero reconstruction loss only when the sparsity constraint holds (or, in these toy models, as sparsity approaches one). If more than m features can be active simultaneously ($a \not\leq m$), the active set exceeds the monosemantic capacity of a single expert; a dense polysemantic model faces the same capacity pressure unless the feature distribution contains exploitable correlations. Therefore, even when $a \not\leq m$ occasionally, the constructed monosemantic MoE may remain competitive under looser sparsity constraints because the dense alternative encounters the same active-feature overload.

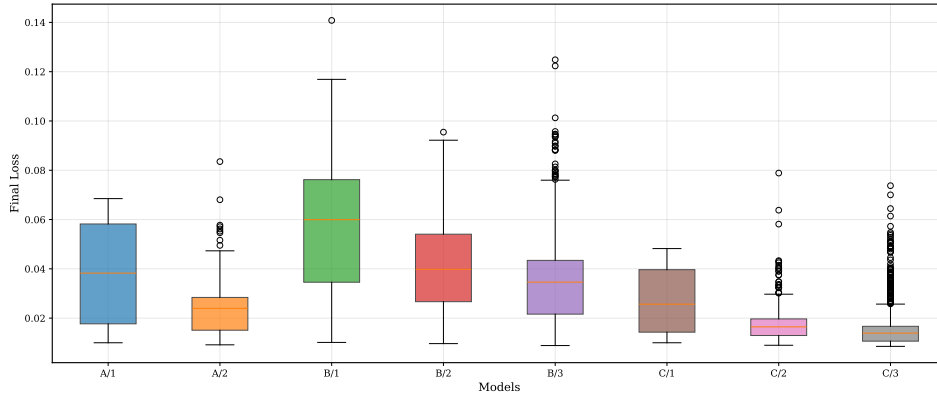


Figure 8: Model X/E uses X for the architecture family from Figure 4 and E for the total number of experts, i.e. network sparsity. Increasing network sparsity lowers mean loss while increasing localized variance, especially when the expert count reaches the number of input feature dimensions. This variance is consistent with less stable MoE training relative to dense models, even though ten models are trained per cell and the lowest-loss run is selected.

Thus, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ operating under $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ such that $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for every valid input. In the toy settings considered in this paper, the sparsity constraint holds in the limit as sparsity goes to one. More broadly, if real data have an upper bound on how many semantically meaningful features must be encoded at once, then a MoE with enough experts and a tractable amount of superposition may be sufficient to encode the relevant features while remaining more interpretable.

A.6 LIMITATIONS

Our results should be interpreted within several boundaries. First, every experiment is conducted in a controlled toy-model setting derived from superposition studies in sparse autoencoders (Elhage et al., 2022). This setting makes interference and monosemanticity directly measurable, but it abstracts away many factors present in large-scale transformers, including multi-step computation, attention, heterogeneous feature distributions, and realistic routing dynamics. Consequently, the extent to which these patterns transfer to deployed MoE architectures remains uncertain.

Second, the architecture and routing choices are deliberately restricted: we use simple experts, fixed top- k routing (often $k = 1$), matched parameter budgets between dense and MoE models, and no auxiliary load-balancing loss except where stated. Practical MoEs often use richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), and they typically optimize tasks beyond reconstruction, such as next-token prediction. These differences may produce qualitatively different representational behavior.

Third, the synthetic feature distributions used here—including their sparsity patterns, feature importances, and independence assumptions—are much simpler than natural data distributions.

Finally, although MoEs show increased monosemanticity and reduced superposition under our fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale. Prior work suggests that these dynamics can change substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

Overall, the study provides mechanistic evidence under clean experimental conditions rather than a direct account of large realistic MoE systems. Validating whether the same scope

810 of claims holds at scale will require methods that extend beyond these toy measurements,
811 since the present metrics do not naturally scale to large models. Developing such scalable
812 analyses is a clear direction for future work.
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